

Sweet bing cherries lower circulating concentrations of markers for chronic inflammatory diseases in healthy humans.

A limited number of studies have demonstrated that some modulators of inflammation can be altered by the consumption of sweet cherries. We have taken a proteomics approach to determine the effects of dietary cherries on targeted gene expression. The purpose was then to determine changes caused by cherry consumption in the plasma concentrations of multiple biomarkers for several chronic inflammatory diseases in healthy humans with modestly elevated C-reactive protein (CRP; range, 1-14 mg/L; mean, 3.5 mg/L; normal, <1.0 mg/L). Eighteen men and women (45-61 y) supplemented their diets with Bing sweet cherries (280 g/d) for 28 d. Fasting blood samples were taken before the start of consuming the cherries (study d 7), 28 d after the initiation of cherry supplementation (d 35), and 28 d after the discontinuation (d 63). Of the 89 biomarkers assessed, cherry consumption for 28 d altered concentrations of 9, did not change those of 67, and the other 13 were below the detection limits. Cherry consumption decreased ($P < 0.05$) plasma concentrations of extracellular newly identified ligand for the receptor for advanced glycation end products (29.0%), CRP (20.1%), ferritin (20.3%), plasminogen activator inhibitor-1 (19.9%), endothelin-1 (13.7%), epidermal growth factor (13.2%), and IL-18 (8.1%) and increased that of IL-1 receptor antagonist (27.9%) compared with corresponding values on study d 7. The ferritin concentration continued to decrease between d 35 and 63 and it was significantly lower on d 63 than on d 7. Because the participants in this study were healthy, no clinical pathology end points were measured. However, results from the present study demonstrate that cherry consumption selectively reduced several biomarkers associated with inflammatory diseases.